Part 1: Definitions and Basic Concepts

1. Define a lens. Name the two main types of spherical lenses.

2. Define the 'Optical Centre' of a lens. What is its special property regarding a ray of light passing through it?

3. Define the First Principal Focus of a convex lens.

4. Define the Second Principal Focus of a concave lens.

5. Define the focal length of a lens. State its SI unit.

6. What is meant by the 'power of a lens'? State its SI unit.

7. How is the power of a lens related to its focal length? Write the mathematical expression.

8. Define the linear magnification produced by a lens.

9. What is the sign convention for the focal length of (i) a convex lens, and (ii) a concave lens?

10. What do you understand by the statement "The power of a given lens is +2.0 D"?

Part 2: Give Reasons & Logical Applications

11. Why is a convex lens called a converging lens while a concave lens is called a diverging lens?

12. A piece of glass immersed in a particular liquid becomes completely invisible. What can you conclude about the refractive indices of the glass and the liquid?

13. How does the focal length of a convex lens change if it is immersed in water instead of being kept in the air? Give a reason.

14. Can a concave lens ever form a real image of a real object? Justify your answer.

15. Which type of lens is used as a simple magnifying glass? Where must the object be placed to achieve this?

16. How does the focal length of a lens depend on the color (wavelength) of incident light? Does a lens have the same focal length for red and blue light?

17. Why is the magnification produced by a concave lens always less than 1 (for a real object)?

18. A convex lens forms a virtual image. How is the position of the object related to the focal length of the lens?

19. You are provided with a printed piece of paper. How will you identify whether a given lens is convex or concave just by looking at the print through it, without touching the lens?

20. What happens to the image formed by a convex lens if its lower half is covered with black paper?

Part 3: Ray Diagrams, Rules, and Comparisons

21. State three rules for drawing ray diagrams for the formation of an image by a convex lens.

22. Differentiate between a real image and a virtual image (state any two key differences).

23. Differentiate between a convex lens and a concave lens based on their physical shape and their action on an incident parallel beam of light.

24. Draw a neat, labelled ray diagram to show the formation of a real, inverted, and magnified image by a convex lens. State the position of the object.

25. Draw a ray diagram to show the formation of an image when an object is placed between the optical centre and the principal focus of a convex lens. State the nature of the image.

26. Show by drawing a ray diagram that a concave lens always forms a virtual, erect, and diminished image of a real object, regardless of the object's position.

27. Where should an object be placed in front of a convex lens so that the image is of the exact same size as the object? Draw the corresponding ray diagram.

28. Draw a ray diagram to show the use of a convex lens as a reading glass (magnifying glass).

29. State the Cartesian sign convention for measuring distances in lenses (specifically regarding object distance and image distance).

30. The magnification of an image formed by a lens is -1.5. What does the negative sign indicate about the nature of the image? Is the lens convex or concave?

Part 4: Expected Numericals

(Use the lens formula $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ and magnification $m=\frac{v}{u}$ where applicable)

31. A convex lens has a focal length of 20 cm. Calculate its power.

32. Two lenses of power +2.5 D and -1.5 D are placed in contact. Find the total power and the focal length of the combination.

33. An object is placed at a distance of 30 cm from a convex lens of focal length 20 cm. Find the position and nature of the image.

34. A concave lens has a focal length of 15 cm. At what distance should an object be placed from the lens so that it forms an image at 10 cm from the lens?

35. Calculate the linear magnification produced by the lens in question 34.

36. A needle 5 cm long is placed on the principal axis of a convex lens of focal length 10 cm. If the object distance is 15 cm, find the size (height) of the image.

37. An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm. Find the position of the image and its magnification.

38. A convex lens forms a real and inverted image of a needle at a distance of 50 cm from it. Where is the needle placed in front of the convex lens if the image is exactly equal to the size of the object? Also, find the power of the lens.

39. A lens forms an upright and magnified image of an object. The magnification is +2. If the object is placed at 10 cm from the lens, calculate the focal length of the lens and identify its type.

40. An object is placed 20 cm in front of a lens of power -5.0 D. Calculate the image distance and state the characteristics of the image formed.